Mathematical Concepts

for (MATH261) Discrete Mathematics

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This document develops the principal mathematical concepts, definitions, and results from the MATH261 lectures. General mathematical notation and writing follow *(MATH000) Mathematical Language*. Problems and worked applications are collected separately in the midterm preparation document.

Theorem 1. Pigeonhole Principle
If N objects are distributed among $k \geq 1$ boxes, at least one box contains at least $\lceil N/k \rceil$ objects. In particular, distributing $k+1$ objects among k boxes puts at least two objects in one box.

Proof. If every box contained fewer than $\lceil N/k \rceil$ objects, each would contain at most $\lceil N/k \rceil - 1 < N/k$ objects. The total would then be less than N, a contradiction. The particular case follows by setting $N = k+1$. □

Definition 2. Binary Relation
A *binary relation* R on a set X is a subset of $X \times X$. We write xRy to mean $(x, y) \in R$. The relation R is

(i) *reflexive* if xRx for every $x \in X$;
(ii) *symmetric* if xRy implies yRx for all $x, y \in X$;
(iii) *antisymmetric* if xRy and yRx together imply $x = y$;
(iv) *transitive* if xRy and yRz together imply xRz.

Symmetry and antisymmetry are different conditions; neither is the negation of the other.

Definition 3. Equivalence Relation and Equivalence Class
An *equivalence relation* $\sim$ on X is reflexive, symmetric, and transitive. For $x \in X$, its *equivalence class* is

$$[x]_\sim \coloneqq \{y \in X : y \sim x\}.$$

Definition 4. Partition
A *partition* of a set X is a family $\mathcal{P}$ of nonempty, pairwise disjoint subsets whose union is X. For $X = \varnothing$, the empty family is its unique partition.

Theorem 5. Equivalence Relations and Partitions
The equivalence classes of an equivalence relation on X form a partition of X. Conversely, every partition $\mathcal{P}$ of X induces an equivalence relation given by $x \sim y$ if and only if x and y belong to the same block of $\mathcal{P}$. These constructions are inverse to each other.

Proof. Reflexivity gives $x \in [x]_\sim$, so the classes cover X and are nonempty. If $z \in [x]_\sim \cap [y]_\sim$, then $x \sim z$ and $z \sim y$ by symmetry, hence $x \sim y$ by transitivity. Every member of $[x]_\sim$ is therefore equivalent to y, and conversely; thus $[x]_\sim = [y]_\sim$. Distinct classes are disjoint.

Conversely, every $x \in X$ lies in exactly one block of a partition. Belonging to the same block is reflexive, symmetric, and transitive. Its equivalence class at x is precisely the block containing x, establishing both inverse assertions. □

Definition 6. Partial Order, Poset, and Total Order
A *partial order* $\preceq$ on X is a reflexive, antisymmetric, and transitive relation. The pair $(X, \preceq)$ is a *partially ordered set* (*poset*). Two elements x, y are *comparable* if $x \preceq y$ or $y \preceq x$; otherwise they are *incomparable*. A partial order is a *total order* if every pair is comparable. We write $x \prec y$ if $x \preceq y$ and $x \neq y$.

Definition 7. Cover Relation and Hasse Diagram
For x, y in a poset, y *covers* x, written $x \lessdot y$, if $x \prec y$ and there is no z with $x \prec z \prec y$. A *Hasse diagram* of a finite poset places its elements as vertices, with higher elements drawn above lower elements, and draws a segment between x and y precisely when $x \lessdot y$. Reflexive loops, arrowheads, and edges implied by transitivity are omitted. The order is recovered from upward paths, including paths of length zero.

Definition 8. Extremal Elements
Let $(X, \preceq)$ be a poset and $A \subseteq X$. An element $m \in A$ is *maximal* in A if no $a \in A$ satisfies $m \prec a$, and *minimal* if no $a \in A$ satisfies $a \prec m$. An element $g \in A$ is *greatest* in A if $a \preceq g$ for every $a \in A$, and *least* if $g \preceq a$ for every $a \in A$.

Proposition 9. Greatest and Maximal Elements
A greatest (respectively, least) element is unique, if it exists, and is maximal (respectively, minimal). A finite nonempty poset has a maximal and a minimal element, but need not have a greatest or a least element.

Proof. If g, h are both greatest, then $g \preceq h$ and $h \preceq g$, so antisymmetry gives $g = h$. If $g \prec a$ for some a, greatestness also gives $a \preceq g$, again contradicting antisymmetry. The dual assertions follow similarly. In a finite nonempty poset, start at any element and move to a strictly greater element whenever possible. Finiteness forces termination at a maximal element. Reversing the order gives a minimal element. □

Definition 10. Sunflower
A family of distinct sets $A_1, \ldots, A_r$ is a *sunflower* with r petals if a set K exists such that $A_i \cap A_j = K$ for every $i \neq j$. The set K is the *core*; the sets $A_i \setminus K$ are pairwise disjoint *petals*. In a k-uniform family every member has cardinality k.

Lemma 11. Sunflower Lemma
Let $k, r \in \mathbb{N}$ and let $\mathcal{F}$ be a family of distinct sets, each containing exactly k elements. If

$$|\mathcal{F}| > k!(r-1)^k,$$

then $\mathcal{F}$ contains a sunflower with r petals.

Proof. Induct on k. For $k = 1$, more than $r - 1$ distinct singleton sets have pairwise empty intersections, so any r form a sunflower. The case $r = 1$ is immediate for every k: the inequality guarantees $\mathcal{F} \neq \varnothing$.

Assume $k \geq 2$ and $r \geq 2$, and take a maximal pairwise disjoint subfamily $\{F_1, \ldots, F_t\} \subseteq \mathcal{F}$. If $t \geq r$, its first r members form a sunflower with empty core. Otherwise $t \leq r - 1$. Put $U = F_1 \cup \cdots \cup F_t$, so $|U| \leq k(r-1)$. Maximality implies that every $F \in \mathcal{F}$ meets U. Counting pairs (u, F) with $u \in U \cap F$ shows that some $u \in U$ belongs to at least

$$\frac{|\mathcal{F}|}{|U|} > \frac{k!(r-1)^k}{k(r-1)} = (k-1)!(r-1)^{k-1}$$

members. Removing u from each such member produces distinct $(k-1)$-element sets. By induction these contain a sunflower with r petals and core K; reinserting u gives the required sunflower with core $K \cup \{u\}$. □

Definition 12. Ordered Selections and Permutations
For $0 \le r \le n$, an *ordered selection without repetition* of r elements from an n-element set is an injective map $[r] \to [n]$ after identifying the ground set with $[n]$. The number of such selections is

$$ {}_nP_r = n^{\underline{r}} = \frac{n!}{(n-r)!}. $$

A *permutation* of a set X is a bijection $X \to X$. In particular, S_n denotes the set of permutations of $[n]$; it has cardinality $n! = {}_nP_n$.

Proof. The first position has n choices, the second has $n-1$, and the rth has $n-r+1$. Multiplication gives $n^{\underline{r}}$. For $r = n$, an injection of the finite set $[n]$ into itself is a bijection. □

Notation 13. Permutation Notation
For $\pi \in S_n$, the expression $\pi(i)$ is the image of i and

$$ \pi = \begin{pmatrix} 1 & 2 & \cdots & n \\ \pi(1) & \pi(2) & \cdots & \pi(n) \end{pmatrix} $$

is its *two-line notation*. The composition $\sigma \circ \pi$ sends i to $\sigma(\pi(i))$: the rightmost permutation acts first. The identity id fixes every element, and π^{-1} denotes the inverse permutation. A *cycle* $(i_1\ i_2\ \cdots\ i_\ell)$ sends i_j to i_{j+1}, with $i_{\ell+1} = i_1$, and fixes elements not listed. Every permutation has a decomposition into disjoint cycles, unique up to the order of cycles and cyclic rotation within each cycle; fixed points are cycles of length one.

Proof of Cycle Decomposition. Starting from any $i \in [n]$, successive images $i, \pi(i), \pi^2(i), \ldots$ must eventually repeat. Injectivity forces the first repeated element to be i, giving a cycle. Remove its elements and repeat with any unused element. The resulting cycles are disjoint and cover $[n]$. Each cycle is the orbit of one of its elements, establishing uniqueness. □

Definition 14. Combinations
An *r-combination* of an n-element set is an r-element subset. Its number is

$$ {}_nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}, \qquad 0 \le r \le n. $$

We use $\binom{n}{r} = 0$ for integers $r < 0$ or $r > n$, when $n \ge 0$.

Proof. Every r-subset has $r!$ orders, so the $n!/(n-r)!$ ordered selections fall into groups of $r!$. Dividing gives the formula. □

Proposition 15. Pascal's Identity
For $n \ge 1$ and $0 \le r \le n$,

$$ \binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}. $$

Proof. Fix x in an n-element set. Every r-subset either contains x, in which case its remaining $r-1$ elements come from the other $n-1$, or does not contain x, in which case all r elements come from those $n-1$. The alternatives are disjoint and exhaustive. The factorial formula yields the same identity algebraically. □

Theorem 16. Binomial Theorem
For $n \in \mathbb{N}_0$ and commuting variables x, y,

$$ (x+y)^n = \sum_{r=0}^{n} \binom{n}{r} x^{n-r} y^r. $$

Proof. In expanding n copies of $(x+y)$, a contribution $x^{n-r}y^r$ is determined by choosing precisely the r factors from which y is taken, giving $\binom{n}{r}$ contributions. Alternatively, the identity holds at $n = 0$; multiplying its nth instance by $x + y$, and combining adjacent powers using Pascal's identity, proves the $(n + 1)$st instance. □

Definition 17. Multinomial Coefficient
For $n_1, \ldots, n_k \in \mathbb{N}_0$ with $n_1 + \cdots + n_k = n$, define

$$\binom{n}{n_1, \ldots, n_k} \coloneqq \frac{n!}{n_1! \cdots n_k!}.$$

It counts assignments of n distinguishable objects to k labeled groups with prescribed sizes $n_1, \ldots, n_k$.

Proof. Order the n objects, then place the first n_1 in group one, the next n_2 in group two, and so on. Every grouping corresponds to exactly $n_1! \cdots n_k!$ orders, one for each ordering within the groups. □

Theorem 18. Multinomial Theorem
For commuting variables $x_1, \ldots, x_k$ and $n \in \mathbb{N}_0$,

$$(x_1 + \cdots + x_k)^n = \sum_{\substack{n_1, \ldots, n_k \geq 0 \\ n_1 + \cdots + n_k = n}} \binom{n}{n_1, \ldots, n_k} \prod_{i=1}^{k} x_i^{n_i}.$$

Proof. An expansion term is specified by choosing one of the k variables from each of the n factors. For fixed multiplicities $n_1, \ldots, n_k$, the number of choices is the corresponding multinomial coefficient, since this is the number of ways to partition the n factor positions into labeled groups of these sizes. Summing over all multiplicities proves the formula. □

Definition 19. Combinations with Repetition
For $n \geq 1$ and $r \in \mathbb{N}_0$, a combination with repetition is a multiset of cardinality r drawn from n types. Its number is denoted

$${}_nH_r = \left(\!\binom{n}{r}\!\right) = \binom{n + r - 1}{r}.$$

Theorem 20. Stars and Bars
For $n \in \mathbb{N}$ and $r \in \mathbb{N}_0$, the number of solutions to

$$x_1 + \cdots + x_n = r, \qquad (x_1, \ldots, x_n) \in \mathbb{N}_0^n,$$

is $\binom{n+r-1}{n-1} = \binom{n+r-1}{r}$. It is also the number ${}_nH_r$ of r-combinations with repetition from n types.

Proof. Represent the solution by x_1 stars, a bar, x_2 stars, and so on, ending with x_n stars. This gives a string of r stars and $n - 1$ bars. Conversely, the numbers of stars between consecutive bars recover the x_i uniquely. Choosing the positions of the $n - 1$ bars among $n + r - 1$ positions gives the formula. Selecting x_i objects of type i is the same data as the solution vector. □

Theorem 21. Inclusion–Exclusion Principle
For finite sets $A_1, \ldots, A_n$,

$$\left| \bigcup_{i=1}^{n} A_i \right| = \sum_{\varnothing \neq I \subseteq [n]} (-1)^{|I|+1} \left| \bigcap_{i \in I} A_i \right|.$$

Equivalently, if $A_i \subseteq U$ for a finite universe U, then

$$\left| U \setminus \bigcup_{i=1}^{n} A_i \right| = \sum_{I \subseteq [n]} (-1)^{|I|} \left| \bigcap_{i \in I} A_i \right|, \qquad \bigcap_{i \in \varnothing} A_i = U.$$

Proof. Fix $u \in U$ that lies in exactly t of the sets. If $t = 0$, its contribution to the first right-hand side is zero. If $t \geq 1$, its contribution is

$$\sum_{j=1}^{t} (-1)^{j+1} \binom{t}{j} = 1 - (1-1)^t = 1.$$

Thus each element of the union is counted exactly once. Subtracting the first identity from $|U|$ proves the complement form.

For a second proof, induct on n. Apply the two-set identity $|X \cup Y| = |X| + |Y| - |X \cap Y|$ with $X = \bigcup_{i=1}^{n-1} A_i$ and $Y = A_n$. The induction hypothesis expands $|X|$; applying it to the sets $A_i \cap A_n$ expands $|X \cap A_n|$. The first sum contains exactly the nonempty index sets not containing n, while the subtracted sum contains those containing n, with the required alternating signs.□

Proposition 22. Number of Surjections
Let $n, k \in \mathbb{N}_0$. For $k \geq 1$, the number of surjections $[n] \to [k]$ is

$$\#\operatorname{Surj}([n],[k]) = \sum_{j=0}^{k} (-1)^j \binom{k}{j} (k-j)^n.$$

Here $0^0 = 1$ when an empty domain is counted. For $n = k = 0$ there is one surjection $\varnothing \to \varnothing$; for $n > 0$ and $k = 0$ there are none.

Proof. For $k \geq 1$, there are k^n maps $[n] \to [k]$. Let A_i be those maps whose image misses i. For any $J \subseteq [k]$, a map missing every element of J has $(k - |J|)^n$ choices. Inclusion–exclusion counts the maps in none of the A_i, which are exactly the surjections. □

Definition 23. Euler's Totient Function
For $n \in \mathbb{N}$, Euler's totient function is

$$\varphi(n) \coloneqq \#\{k \in [n] : \gcd(k,n) = 1\}.$$

In particular, $\varphi(1) = 1$.

Theorem 24. Euler's Totient Formula
If $n \in \mathbb{N}$ has distinct prime divisors $p_1, \ldots, p_t$, then

$$\varphi(n) = n \prod_{i=1}^{t} \left(1 - \frac{1}{p_i}\right) = n \prod_{\substack{p \mid n \\ p \text{ prime}}} \left(1 - \frac{1}{p}\right).$$

The empty product for $n = 1$ equals 1.

Proof. For $n \geq 2$, an integer in $[n]$ fails to be coprime to n precisely when it is divisible by at least one of $p_1, \ldots, p_t$. For a subset $I \subseteq [t]$, exactly $n / \prod_{i \in I} p_i$ integers in $[n]$ are divisible by all the primes indexed by I. By inclusion–exclusion, the number divisible by none is

$$\sum_{I \subseteq [t]} (-1)^{|I|} \frac{n}{\prod_{i \in I} p_i} = n \prod_{i=1}^{t} \left(1 - \frac{1}{p_i}\right).$$

For $n = 1$, the definition and empty-product convention agree. □

Definition 25. Derangements
A *derangement* of $[n]$ is a permutation $\pi \in S_n$ with no fixed point, so $\pi(i) \neq i$ for every $i \in [n]$. Denote the number of derangements by D_n (also $!n$). The empty permutation is a derangement by convention, so $D_0 = 1$; furthermore, $D_1 = 0$ and $D_2 = 1$.

Theorem 26. Derangement Formula
For every $n \in \mathbb{N}_0$,

$$D_n = n! \sum_{j=0}^{n} \frac{(-1)^j}{j!}.$$

Proof. For each $i \in [n]$, let $A_i = \{\pi \in S_n : \pi(i) = i\}$. If $I \subseteq [n]$ has cardinality j, fixing all elements of I leaves $(n-j)!$ permutations. Inclusion–exclusion therefore gives

$$D_n = \sum_{j=0}^{n} (-1)^j \binom{n}{j} (n-j)! = n! \sum_{j=0}^{n} \frac{(-1)^j}{j!}.$$

The formula also gives $D_0 = 1$. □

Theorem 27. Derangement Recurrences
For $n \geq 1$ and $n \geq 2$, respectively,

$$D_n = nD_{n-1} + (-1)^n, \qquad D_n = (n-1)(D_{n-1} + D_{n-2}),$$

with $D_0 = 1$ and $D_1 = 0$.

Proof. Subtract the formulas for D_n and nD_{n-1} in Theorem 26 to obtain

$$D_n - nD_{n-1} = n! \frac{(-1)^n}{n!} = (-1)^n.$$

Applying this identity at n and $n-1$ yields

$$\begin{aligned} D_n &= nD_{n-1} + (-1)^n \\ &= (n-1)D_{n-1} + \big((n-1)D_{n-2} + (-1)^{n-1}\big) + (-1)^n \\ &= (n-1)(D_{n-1} + D_{n-2}). \end{aligned}$$

For a combinatorial proof of the second recurrence, choose the image $j = \pi(n)$ of n ($n-1$ choices). If $\pi(j) = n$, deleting the two-cycle $(n\ j)$ leaves a derangement of the other $n-2$ elements. Otherwise, delete n from the cycle containing it, producing a derangement of the remaining $n-1$ elements. These constructions are reversible, giving $(n-1)(D_{n-2} + D_{n-1})$. □

Definition 28. Catalan Numbers
The *Catalan numbers* are

$$C_n \coloneqq \frac{1}{n+1} \binom{2n}{n}, \qquad n \in \mathbb{N}_0.$$

Thus $C_0 = 1$, $C_1 = 1$, $C_2 = 2$, $C_3 = 5$, and $C_4 = 14$.

Theorem 29. Catalan Words
For $n \in \mathbb{N}_0$, the number of words $a_1 \cdots a_{2n}$ with $a_i \in \{-1, 1\}$, exactly n entries of each sign, and

$$\sum_{j=1}^{i} a_j \geq 0 \qquad (1 \leq i \leq 2n)$$

is C_n.

Proof. The assertion for $n = 0$ concerns the single empty word. Let $n \geq 1$. There are $\binom{2n}{n}$ words with n entries of each sign. Call a word *bad* if some prefix sum is negative. Locate the first prefix whose sum is -1 and reverse the signs of all entries in that prefix. Since that prefix contains one more -1 than $+1$, the resulting word has $n+1$ positive entries and $n-1$ negative entries.

This map is a bijection: a word with $n+1$ positive entries first reaches prefix sum $+1$ at a unique position, and reversing the signs up to that position reconstructs a bad word. Hence the number of bad words is $\binom{2n}{n+1}$, and the desired number is

$$\binom{2n}{n} - \binom{2n}{n+1} = \left(1 - \frac{n}{n+1}\right)\binom{2n}{n} = C_n.$$

□

Corollary 30. Catalan Lattice Paths
The number of paths from $(0,0)$ to (n,n) consisting of n right steps and n upward steps that never pass above the diagonal $y = x$ is C_n.

Proof. Replace each right step by $+1$ and each upward step by -1. At every prefix, the partial sum equals the current horizontal coordinate minus the vertical coordinate. Thus remaining on or below the diagonal is equivalent to the condition in Theorem 29. □

Remark. The same word can be read as a balanced parenthesis string, with $+1$ an opening parenthesis and -1 a closing parenthesis. Nonnegative prefix sums guarantee that no closing parenthesis occurs before its matching opening parenthesis. This is a bijective interpretation, not a separate counting problem.

Theorem 31. Catalan Recurrence
With $C_0 = 1$, for every $n \geq 1$,

$$C_n = \sum_{j=0}^{n-1} C_j C_{n-1-j}.$$

Proof. Consider a nonnegative balanced ± 1 word of length $2n$. Let its first return to partial sum zero occur after $2(j+1)$ entries, where $0 \leq j \leq n-1$. Its initial block is a $+1$, followed by a nonnegative balanced word of length $2j$, followed by a -1. The remaining suffix is independently a nonnegative balanced word of length $2(n-1-j)$. For this fixed j there are $C_j C_{n-1-j}$ choices by Theorem 29. Summing over the unique first-return index proves the recurrence. □

Definition 32. Stirling Numbers of the First Kind
Let $c(n,k)$ (also $\left[{n \atop k}\right]$) be the number of permutations in S_n whose disjoint-cycle decomposition has exactly k cycles. This is the *unsigned Stirling number of the first kind*. The *signed Stirling number of the first kind* is

$$s(n,k) \coloneqq (-1)^{n-k} c(n,k).$$

The conventions are $c(0,0) = s(0,0) = 1$, $c(n,0) = 0$ for $n \geq 1$, and $c(n,k) = 0$ for $k > n$. In particular, $c(n,n) = 1$, $c(n,1) = (n-1)!$ for $n \geq 1$, and $c(n,n-1) = \binom{n}{2}$ for $n \geq 2$.

Proposition 33. First-Kind Stirling Recurrences
For $n \geq 1$ and $1 \leq k \leq n$,

$$c(n,k) = c(n-1,k-1) + (n-1)c(n-1,k),$$
$$s(n,k) = s(n-1,k-1) - (n-1)s(n-1,k).$$

Proof. In a permutation of $[n]$ with k cycles, either n is a singleton cycle or it is not. In the first case, removing (n) leaves a permutation of $[n-1]$ with $k-1$ cycles. In the second, remove n from its cycle. The remaining permutation has k cycles, and n can be reinserted immediately after any one of the $n-1$ other elements, giving $n-1$ possibilities. This proves the first recurrence. Multiplying it by $(-1)^{n-k}$ and using $s(n,k) = (-1)^{n-k}c(n,k)$ gives the second. □

Proposition 34. Falling-Factorial Coefficients
For $n \in \mathbb{N}_0$,

$$x^{\underline{n}} = x(x-1)\cdots(x-n+1) = \sum_{k=0}^{n} s(n,k)x^k,$$

with $x^{\underline{0}} = 1$.

Proof. The result is immediate for $n = 0$. If it holds for $n-1$, then

$$x^{\underline{n}} = (x-n+1)\sum_{k=0}^{n-1} s(n-1,k)x^k.$$

The coefficient of x^k on the right is $s(n-1,k-1) - (n-1)s(n-1,k) = s(n,k)$ by Proposition 33. □

Definition 35. Stirling Numbers of the Second Kind
The *Stirling number of the second kind* $S(n,k)$ (also $\left\{ {n \atop k} \right\}$) is the number of partitions of an n-element set into exactly k nonempty, unlabeled blocks. We set

$$S(0,0) = 1, \quad S(n,0) = 0 \ (n \geq 1), \quad S(n,k) = 0 \ (k > n).$$

For $n \geq 1$, $S(n,1) = S(n,n) = 1$; for $n \geq 2$, $S(n,n-1) = \binom{n}{2}$.

Proof of the Special Value. A partition of $[n]$ into $n-1$ nonempty blocks has precisely one block of size two and $n-2$ singleton blocks. Choosing the pair determines the partition and yields $\binom{n}{2}$ choices. □

Theorem 36. Second-Kind Stirling Recurrence
For $n \geq 1$ and $1 \leq k \leq n$,

$$S(n,k) = S(n-1,k-1) + kS(n-1,k).$$

Proof. Consider the block containing n. If it is the singleton $\{n\}$, the other $n-1$ elements form $k-1$ blocks. Otherwise, first partition $[n-1]$ into k blocks and then insert n into exactly one of these k blocks. The two cases are disjoint and exhaustive. □

Theorem 37. Surjections and Stirling Numbers
For $n \in \mathbb{N}_0$ and $k \in \mathbb{N}$,

$$\#\operatorname{Surj}([n],[k]) = k!S(n,k), \qquad S(n,k) = \frac{1}{k!}\sum_{j=0}^{k}(-1)^j\binom{k}{j}(k-j)^n.$$

For $(n,k) = (0,0)$, the corresponding cardinality identity is $1 = 0!S(0,0)$.

Proof. The nonempty fibers of a surjection $f : [n] \to [k]$ form a partition of $[n]$ into k blocks. Conversely, labeling the blocks of such a partition bijectively by $[k]$ determines a surjection; there are $k!$ such labelings. Combining this counting identity with Proposition 22 gives the explicit sum. □

Remark. The identities in Proposition 33 and Theorem 36 have similar forms but count different objects: cycles of permutations versus blocks of set partitions. Their initial conditions and signs must not be interchanged.

Definition 38. Bell Numbers
The *Bell number* B_n counts all partitions of an n-element set. Hence

$$B_n = \sum_{k=0}^{n} S(n,k), \qquad B_0 = 1.$$

The first values are $B_1 = 1$, $B_2 = 2$, $B_3 = 5$, $B_4 = 15$, and $B_5 = 52$.

Theorem 39. Bell Recurrence
For $n \geq 1$,

$$B_n = \sum_{k=0}^{n-1} \binom{n-1}{k} B_k.$$

Proof. Fix one distinguished element x of an n-element set. If exactly $n-1-k$ of the remaining elements lie in the block containing x, choose them in $\binom{n-1}{n-1-k} = \binom{n-1}{k}$ ways. The other k elements may be partitioned in B_k ways. Every partition determines a unique k, so summing proves the recurrence. □

Theorem 40. Bell Finite-Sum Formula
For $n \in \mathbb{N}_0$,

$$B_n = \sum_{k=0}^{n} \frac{k^n}{k!} \left(\sum_{j=0}^{n-k} \frac{(-1)^j}{j!} \right),$$

where $0^0 = 1$ for the term occurring at $n = k = 0$.

Proof. For $n \geq 1$, use the explicit formula in Theorem 37 and rearrange the finite sums:

$$\begin{aligned} B_n &= \sum_{m=0}^{n} \frac{1}{m!} \sum_{j=0}^{m} (-1)^j \binom{m}{j} (m-j)^n \\ &= \sum_{m=0}^{n} \sum_{k=0}^{m} \frac{(-1)^{m-k} k^n}{(m-k)!k!} \\ &= \sum_{k=0}^{n} \frac{k^n}{k!} \sum_{j=0}^{n-k} \frac{(-1)^j}{j!}. \end{aligned}$$

For $n = 0$, both sides equal 1. □

Definition 41. Linear Homogeneous Recurrence Relation
A *linear homogeneous recurrence relation of degree k with constant coefficients* is

$$a_n = c_1 a_{n-1} + \cdots + c_k a_{n-k}, \qquad n \geq k,$$

where $k \geq 1$, $c_1, \ldots, c_k \in \mathbb{R}$, and $c_k \neq 0$. A *solution* is a sequence satisfying the relation for every indicated n.

Proposition 42. Linearity of Homogeneous Solutions
If (a_n) and (b_n) solve the same linear homogeneous recurrence, then so does $(\alpha a_n + \beta b_n)$ for any constants $\alpha, \beta \in \mathbb{R}$.

Proof. For each $n \geq k$, multiply the recurrence for a_n by α, the recurrence for b_n by β, and add. The result is

$$\alpha a_n + \beta b_n = \sum_{j=1}^{k} c_j (\alpha a_{n-j} + \beta b_{n-j}).$$

□

Proposition 43. Existence and Uniqueness from Initial Values
For any prescribed $a_0, \ldots, a_{k-1}$, a recurrence of degree k as above determines exactly one sequence $(a_n)_{n \geq 0}$.

Proof. The recurrence determines a_k from the initial values, then a_{k+1} from the preceding k terms, and so on, establishing existence. If two solutions share the initial terms, induction on n shows that the recurrence forces their remaining terms to agree. □

Definition 44. Characteristic Equation
The *characteristic polynomial* of the recurrence $a_n = \sum_{j=1}^{k} c_j a_{n-j}$ is

$$p(r) \coloneqq r^k - c_1 r^{k-1} - \cdots - c_{k-1} r - c_k.$$

The equation $p(r) = 0$ is its *characteristic equation*. If $r \neq 0$ satisfies this equation, then (r^n) solves the recurrence, since substituting $a_n = r^n$ reduces the recurrence to $r^{n-k} p(r) = 0$.

Theorem 45. Second-Order Recurrences with Distinct Roots
Suppose $c_2 \neq 0$ and $r^2 - c_1 r - c_2$ has two distinct roots $r_1, r_2 \in \mathbb{C}$. Then the solutions of

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} \qquad (n \geq 2)$$

are precisely

$$a_n = \alpha r_1^n + \beta r_2^n, \qquad \alpha, \beta \in \mathbb{C}.$$

For real initial values, the resulting sequence is real; nonreal roots occur in conjugate pairs.

Proof. Each (r_i^n) is a solution by the characteristic equation, and linearity gives every displayed combination. Conversely, given a_0, a_1, the equations

$$\alpha + \beta = a_0, \qquad \alpha r_1 + \beta r_2 = a_1$$

have the unique solution

$$\alpha = \frac{a_1 - a_0 r_2}{r_1 - r_2}, \qquad \beta = \frac{a_0 r_1 - a_1}{r_1 - r_2}.$$

The resulting combination agrees with the prescribed initial values, hence with the original solution by Proposition 43. □

Theorem 46. Second-Order Recurrences with a Repeated Root
Suppose $c_2 \neq 0$ and $r^2 - c_1 r - c_2 = (r - r_0)^2$. Then all solutions of the associated second-order recurrence are

$$a_n = (\alpha + \beta n) r_0^n, \qquad \alpha, \beta \in \mathbb{C}.$$

Proof. Since $r_0 \neq 0$, coefficient comparison gives $c_1 = 2r_0$ and $c_2 = -r_0^2$. Besides r_0^n, the sequence $n r_0^n$ is a solution, because

$$2r_0(n-1) r_0^{n-1} - r_0^2 (n-2) r_0^{n-2} = n r_0^n.$$

By linearity, every displayed combination is a solution. To match arbitrary initial values a_0, a_1, choose $\alpha = a_0$ and $\beta = a_1/r_0 - a_0$. Uniqueness of the recurrence then establishes that every solution has this form. □

Theorem 47. General Solution of a Constant-Coefficient Homogeneous Recurrence
Let $p(r) = r^k - c_1 r^{k-1} - \cdots - c_k$, where $c_k \neq 0$. Suppose its distinct complex roots are $r_1, \ldots, r_t$ with multiplicities $m_1, \ldots, m_t$, so $m_1 + \cdots + m_t = k$. The solutions of the recurrence are exactly

$$a_n = \sum_{i=1}^{t} P_i(n) r_i^n, \qquad \deg P_i < m_i,$$

where each P_i is a polynomial over $\mathbb{C}$. The k polynomial coefficients are uniquely determined by k initial values. For real recurrence coefficients and real initial values, the expression can be chosen in conjugate pairs to yield a real-valued sequence.

Proof. Let E be the forward shift operator on sequences, $(Ea)_n = a_{n+1}$, and let $\Delta P(n) = P(n+1) - P(n)$. The recurrence is equivalent to $p(E)a = 0$ (with the equation indexed starting at $n = 0$). For $r \neq 0$,

$$(E - r)\big(P(n)r^n\big) = r^{n+1}\Delta P(n).$$

Consequently, $(E-r)^m(P(n)r^n) = r^{n+m}\Delta^m P(n) = 0$ whenever $\deg P < m$. Since the factors of $p(E) = \prod_{i=1}^{t}(E - r_i)^{m_i}$ commute, every sequence in the displayed form is a solution.

These k basic solutions are linearly independent. Indeed, suppose $\sum_i P_i(n)r_i^n = 0$ for all $n \geq 0$, with $\deg P_i < m_i$. Fix i and apply $\prod_{j\neq i}(E - r_j)^{m_j}$. All terms except the ith disappear. On sequences $P(n)r_i^n$, each factor $E - r_j$, for $j \neq i$, transforms P into

$$(r_i - r_j)P + r_i\Delta P.$$

On the vector space of polynomials of degree less than m_i, this map is triangular in the monomial basis with nonzero diagonal $r_i - r_j$, hence invertible. Therefore the surviving term can vanish only if $P_i = 0$. As i was arbitrary, all P_i vanish.

Thus the k basic solutions have independent initial-value vectors $(a_0, \ldots, a_{k-1}) \in \mathbb{C}^k$: otherwise their linear combination would have k zero initial terms and hence be identically zero by Proposition 43. These k vectors form a basis of $\mathbb{C}^k$, so every choice of initial values determines exactly one linear combination of the proposed form. □

Definition 48. Linear Nonhomogeneous Recurrence Relation
A *linear nonhomogeneous recurrence relation of degree k with constant coefficients* is

$$a_n = c_1 a_{n-1} + \cdots + c_k a_{n-k} + F(n), \qquad n \geq k,$$

where the c_j are constants and F is a prescribed function. The recurrence obtained by setting $F = 0$ is the *associated homogeneous recurrence*. A *particular solution* is any one sequence satisfying the nonhomogeneous recurrence.

Theorem 49. Homogeneous Plus Particular Solution
If $a_n^{(p)}$ is a particular solution to a linear nonhomogeneous recurrence, then every solution is of the form

$$a_n = a_n^{(p)} + a_n^{(h)},$$

where $a_n^{(h)}$ is an arbitrary solution of the associated homogeneous recurrence.

Proof. Subtract the recurrence satisfied by $a_n^{(p)}$ from the recurrence satisfied by any other solution a_n. Their difference $b_n = a_n - a_n^{(p)}$ satisfies $b_n = \sum_{j=1}^{k} c_j b_{n-j}$ because the forcing terms cancel. Conversely, adding a homogeneous solution to the particular solution preserves the forcing term $F(n)$, yielding another solution. □

Proposition 50. Method of Undetermined Coefficients
Let $p(r)$ be the characteristic polynomial of a linear recurrence with constant coefficients, and suppose its forcing term is $F(n) = P_d(n)\lambda^n$, where $\lambda \in \mathbb{C} \setminus \{0\}$ and P_d is a polynomial of degree at most d. If λ is a root of p of multiplicity m (and $m = 0$ if it is not a root), a particular solution may be sought in the form

$$a_n^{(p)} = n^m Q_d(n)\lambda^n,$$

where Q_d is a polynomial of degree at most d with undetermined coefficients. For a sum of such forcing terms, particular solutions may be added.

Proof. Writing the recurrence using the shift operator as $p(E)a = F$ and substituting $a_n = \lambda^n Q(n)$ shows that

$$p(E)(\lambda^n Q(n)) = \lambda^n p(\lambda(I + \Delta))Q(n).$$

If λ has multiplicity m, then $p(\lambda(1+z)) = z^m h(z)$ with $h(0) \neq 0$. On polynomials, the operator $h(\Delta)$ is invertible because Δ lowers degree and $h(0) \neq 0$. Moreover, Δ^m maps polynomials of the form $n^m Q_d(n)$, $\deg Q_d \leq d$, bijectively onto polynomials of degree at most d: its action on monomials is triangular with nonzero leading coefficients. Hence coefficients of Q_d can be chosen uniquely to match P_d. Superposition follows from linearity. □

Forcing term $F(n)$	Particular-solution trial
$P_d(n)$	$n^m Q_d(n)$, where m is the multiplicity of 1 as a characteristic root
λ^n $(\lambda \neq 0)$	$n^m C\lambda^n$, where m is the multiplicity of λ
$P_d(n)\lambda^n$ $(\lambda \neq 0)$	$n^m Q_d(n)\lambda^n$
Sum of terms of these forms	Add the corresponding particular-solution trials

Remark. The method specifies a finite-dimensional trial family, not a guess without justification. After finding its coefficients, add the general homogeneous solution from Theorem 47 and use the initial conditions. Here P_d is prescribed, whereas Q_d has coefficients to be determined. A constant or polynomial forcing term corresponds to $\lambda = 1$.

Definition 51. Linear Recurrences with Variable Coefficients
A *linear recurrence with variable coefficients* has the form

$$a_n = \sum_{j=1}^{k} c_j(n) a_{n-j} + F(n),$$

with prescribed coefficient functions $c_j(n)$. In contrast to constant-coefficient recurrences, a single characteristic polynomial generally does not determine the solutions. A useful *reduction strategy* is to introduce a transformed sequence $b_n = u(n)a_n$ so that the recurrence for b_n has simpler, preferably constant, coefficients.

Proposition 52. First-Order Variable-Coefficient Recurrences
Let

$$a_{n+1} = f(n)a_n + g(n), \qquad n \in \mathbb{N}_0,$$

where a_0 is given. With the convention that empty products equal 1, the unique solution is

$$\boxed{a_n = a_0 \prod_{j=0}^{n-1} f(j) + \sum_{m=0}^{n-1} g(m) \prod_{j=m+1}^{n-1} f(j).}$$

This formula remains valid even if some $f(j) = 0$. If $f(j) \neq 0$ for $0 \leq j < n$, it can also be written as

$$a_n = \left(\prod_{j=0}^{n-1} f(j) \right) \left(a_0 + \sum_{m=0}^{n-1} \frac{g(m)}{\prod_{j=0}^{m} f(j)} \right).$$

Proof. For $n = 0$, the first formula gives a_0. Suppose it holds at n. Multiplying by $f(n)$ and adding $g(n)$ yields

$$\begin{aligned} a_{n+1} &= a_0 \prod_{j=0}^{n} f(j) + \sum_{m=0}^{n-1} g(m) \prod_{j=m+1}^{n} f(j) + g(n) \\ &= a_0 \prod_{j=0}^{n} f(j) + \sum_{m=0}^{n} g(m) \prod_{j=m+1}^{n} f(j), \end{aligned}$$

which is the same formula at $n+1$. Hence it holds by induction, and uniqueness follows from forward recursion. When the factors are nonzero, factor out $\prod_{j=0}^{n-1} f(j)$ to obtain the second expression. Equivalently, divide the recurrence by the cumulative product and telescope. □

Remark. The product–sum formula is preferable to a quotient formula when some coefficient vanishes. In particular, a recurrence with $f(0) = 0$ must not be solved by dividing by $f(0)$; one may use the first formula or begin the quotient argument after the exceptional index.

Definition 53. Ordinary Generating Function
For a sequence $(a_n)_{n\geq 0}$ over $\mathbb{R}$ or $\mathbb{C}$, its *ordinary generating function* (OGF) is the formal power series

$$A(x) \coloneqq \sum_{n=0}^{\infty} a_n x^n.$$

The notation $[x^n]A(x)$ denotes the coefficient a_n. Unless convergence is explicitly asserted, an identity of generating functions is an identity of formal power series, and does not require assigning a numerical value to x.

Proposition 54. Ordinary Generating Function Operations
Let $A(x) = \sum_{n\geq 0} a_n x^n$ and $B(x) = \sum_{n\geq 0} b_n x^n$. Then

$$A'(x) = \sum_{n\geq 0} (n+1)a_{n+1}x^n,$$
$$A(x) + B(x) = \sum_{n\geq 0} (a_n + b_n)x^n,$$
$$A(x)B(x) = \sum_{n\geq 0} \left(\sum_{j=0}^{n} a_j b_{n-j} \right) x^n.$$

In particular, $[x^n](AB) = \sum_{j=0}^{n} a_j b_{n-j}$ is the Cauchy product *formula.*

Proof. Formal differentiation gives $\sum_{n\geq 1} n a_n x^{n-1}$, and reindexing yields the first equality. Addition is coefficientwise. In the product, a monomial x^n arises precisely by multiplying $a_j x^j$ by $b_{n-j}x^{n-j}$ for $j = 0, \ldots, n$, proving the third equality. Each coefficient of a product involves only finitely many summands, so no convergence assumption is needed. □

Definition 55. Generalized Binomial Coefficient
For $\alpha \in \mathbb{R}$ and $n \in \mathbb{N}_0$, define

$$\binom{\alpha}{0} \coloneqq 1, \qquad \binom{\alpha}{n} \coloneqq \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!} = \frac{\alpha^{\underline{n}}}{n!} \quad (n \geq 1).$$

When α is a nonnegative integer, this agrees with the usual binomial coefficient and vanishes for $n > \alpha$.

Theorem 56. Generalized Binomial Theorem
For every $\alpha \in \mathbb{R}$,

$$(1+x)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n \qquad (|x| < 1).$$

The coefficient identity also holds as the formal binomial expansion of $(1+x)^\alpha$. For nonnegative integer α, the series terminates and is the ordinary Binomial Theorem.

Proof. Write $b_n = \binom{\alpha}{n}$ and $F(x) = \sum_{n\geq 0} b_n x^n$. The coefficients satisfy

$$(n+1)b_{n+1} = (\alpha - n)b_n.$$

The ratio test shows that F converges absolutely for $|x| < 1$ unless it terminates, in which case convergence is automatic. On this interval, termwise differentiation is valid. Comparing coefficients using the displayed recurrence gives

$$(1+x)F'(x) = \alpha F(x), \qquad F(0) = 1.$$

Since $1 + x > 0$ for $-1 < x < 1$,

$$\frac{d}{dx}\left((1+x)^{-\alpha}F(x)\right) = 0.$$

The quantity in parentheses is therefore constant and equals its value 1 at $x = 0$. Thus $F(x) = (1+x)^{\alpha}$ throughout $|x| < 1$. The coefficient recurrence also verifies the formal power-series identity, independently of convergence. □

Remark. For $m \in \mathbb{N}$, negative exponents give $\binom{-m}{n} = (-1)^n\binom{m+n-1}{n}$. The corresponding explicit expansions of $(1+x)^{-m}$ and $(1-x)^{-m}$ may be obtained from Theorem 56; their derivation as worked examples belongs in the midterm document.

Definition 57. Exponential Generating Function
For a sequence $(a_n)_{n\geq 0}$, its *exponential generating function* (EGF) is

$$\widehat{A}(x) \coloneqq \sum_{n=0}^{\infty} a_n \frac{x^n}{n!}.$$

Its coefficients recover the original sequence by $a_n = n![x^n]\widehat{A}(x)$. As for OGFs, formal power-series manipulations do not require convergence.

Proposition 58. Exponential Generating Function Operations
Let $\widehat{A}(x) = \sum_{n\geq 0} a_n x^n/n!$ *and* $\widehat{B}(x) = \sum_{n\geq 0} b_n x^n/n!$. *Then*

$$\widehat{A}'(x) = \sum_{n\geq 0} a_{n+1}\frac{x^n}{n!},$$

$$\widehat{A}(x) + \widehat{B}(x) = \sum_{n\geq 0}(a_n + b_n)\frac{x^n}{n!},$$

$$\widehat{A}(x)\widehat{B}(x) = \sum_{n\geq 0}\left(\sum_{j=0}^{n}\binom{n}{j}a_j b_{n-j}\right)\frac{x^n}{n!}.$$

Proof. Differentiation of $a_n x^n/n!$ gives $a_n x^{n-1}/(n-1)!$ for $n \geq 1$, yielding the first identity after reindexing. The second follows coefficientwise. For the third, the coefficient of x^n in the product is

$$\sum_{j=0}^{n}\frac{a_j}{j!}\frac{b_{n-j}}{(n-j)!} = \frac{1}{n!}\sum_{j=0}^{n}\binom{n}{j}a_j b_{n-j},$$

which gives the asserted EGF coefficient formula. □

Remark. The product rules for ordinary and exponential generating functions differ: ordinary products involve the convolution $\sum_j a_j b_{n-j}$, whereas exponential products involve the binomially weighted convolution $\sum_j \binom{n}{j}a_j b_{n-j}$. Deriving the generating functions of Catalan, derangement, Bell, and Stirling sequences constitutes separate worked applications of these rules.